

AWS Certified Developer Associate - Practice Problems 1

AWS Developer Certification Practice Questions

1. A programmer is developing a Node.js application that will be run on a Linux server in their on-premises data center. The application will access various AWS services such as S3, DynamoDB, and ElastiCache using the AWS SDK.

Which of the following is the MOST suitable way to provide access for the developer to accomplish the specified task?

- a. Create an IAM role with the appropriate permissions to access the required AWS services. Assign the role to the on-premises Linux server.
- b. Go to the AWS Console and create a new IAM user with programmatic access. In the application server, create the credentials file at `~/.aws/credentials` with the access keys of the IAM user.
- c. Create an IAM role with the appropriate permissions to access the required AWS services and assign the role to the on-premises Linux server. Whenever the application needs to access any AWS services, request temporary security credentials from STS using the AssumeRole API.
- d. Go to the AWS Console and create a new IAM User with the appropriate permissions. In the application server, create the credentials file at `~/.aws/credentials` with the username and the hashed password of the IAM User.

Answer (b)

2. A developer is moving a legacy web application from their on-premises data center to AWS. The application is used simultaneously by thousands of users, and their session states are stored in memory. The on-premises server usually reaches 100% CPU Utilization every time there is a surge in the number of people accessing the application.

Which of the following is the best way to re-factor the performance and availability of the application's session management once it is migrated to AWS?

- a. Use an ElastiCache for Redis cluster to store the user session state of the application.
- b. Store the user session state of the application using CloudFront.
- c. Use an ElastiCache for Memcached cluster to store the user session state of the application.
- d. Use Sticky Sessions with Local Session Caching.

Answer (a)

3. A company stores confidential data on an Amazon Simple Storage Service (S3) bucket. New regulatory guidelines require that files be stored with server-side encryption. The encryption used must be Advanced Encryption Standard (AES-256) and the company does not want to manage S3 encryption keys.

Which of the following options should you use?

AWS Certified Developer Associate - Practice Problems 2

4. A company is looking at storing their less frequently accessed files on AWS that can be concurrently accessed by hundreds of EC2 instances. The company needs the most cost-effective file storage service that provides immediate access to data whenever needed.

Which of the following options represents the best solution for the given requirements?

- a. Amazon Elastic File System (EFS) Standard IA Storage Class
- b. Amazon S3 Standard Infrequent Access (S3 Standard IA) Storage Class
- c. Amazon Elastic Block Store (EBS)
- d. Amazon Elastic File System (EFS) Standard Storage Class

Answer (a) - Amazon Elastic File System Standard IA Storage Class - a file storage service for use with Amazon Compute (EC2, container, serverless) and on premise servers. The Standard IA Storage Class reduces storage costs for files that are not accessed every day.

5. A company's e-commerce application becomes slow when traffic spikes. The application has a three-tier architecture (web, application and database tier) that uses synchronous transactions. The development team at the company has identified certain bottlenecks in the application tier and it is looking for a long term solution to improve the application's performance.

As a developer associate, which of the following solutions would you suggest to meet the required application response times while accounting for any traffic spikes?

- a. Leverage horizontal scaling for the application's persistence layer by adding Oracle RAC on AWS
- b. Leverage SQS with asynchronous AWS Lambda calls to decouple the application and data tiers
- c. Leverage vertical scaling for the application instances by provisioning a larger Amazon EC2 instance size
- d. Leverage horizontal scaling for the web and application tiers by using Auto Scaling Groups and Application Load Balancer

Answer (d) - A horizontally scalable system is one that can increase capacity by adding more computers to the system. The option will require fewer design changes, mostly configuration changes and the ability for the web/application tier to be able to communicate across instances.

6. A junior developer working on ECS instances terminated a container instance in Amazon Elastic Container Service (Amazon ECS) as per instructions from the team lead. But the container instance continues to appear as a resource in the ECS cluster.

As a Developer Associate, which of the following solutions would you recommend to fix this behavior?

- a. A custom software on the container instance could have failed and resulted in the container hanging in an unhealthy state till restarted again
- b. The container instance has been terminated with AWS CLI, whereas, for ECS instances, Amazon ECS CLI should be used to avoid any synchronization issues
- c. You terminated the container instance while it was in RUNNING state, that lead to this synchronization issues
- d. You terminated the container instance while it was in STOPPED state, that lead to this synchronization issues

Answer - (d) If you terminate a container instance while it is in the STOPPED state, the container instance is not automatically removed from the cluster. Developers will need to deregister the container instance in the STOPPED state by using ECS console or AWS CLI. The container instance will no longer appear as a resource after being deregistered.

7. A company follows collaborative development practices. The engineering manager wants to isolate the development effort by setting up simulations of API components owned by various development teams.

Which API integration type is best suited for this requirement?

- a. AWS_PROXY
- b. MOCK
- c. HTTP_PROXY
- d. HTTP

Answer - (b) MOCK - A collaborative development, a team can isolate their development effort by setting up simulations of API components owned by other teams by using the MOCK integration

8. You are running a website on 10 EC2 instances fronted by an Elastic Load Balancer. Your users are complaining about the fact that the website always asks them to re-authenticate when they are moving between website pages. You are puzzled because it's working just fine on your machine and in the Dev environment with 1 EC2 instance. What could be the reason?
- The Elastic Load Balancer does not have Sticky Sessions enabled
 - The website must have an issue when hosted on multiple EC2 instance
 - The EC2 Instance logs out users as they cannot see their IP addresses instead they receive ELB IP Address

Answer (a) - ELB Sticky Session feature ensure traffic for the same client is always redirected to the same target (e.g. EC2 Instances)

9. You are using an Application Load Balancer to distribute traffic to your website hosted on EC2 instances. It turns out that your website only sees traffic coming from private IPv4 addresses which are in fact your Application Load Balancer's IP addresses. What should you do to get the IP address of clients connected to your website?
- Modify the website

10. You hosted an application on a set of EC2 instances fronted by an Elastic Load Balancer. A week later, users begin complaining that sometimes the application just doesn't work. You investigate the issue and found that some EC2 instances crash from time to time. What should you do to protect users from connecting to the EC2 instances that are crashing?
- Enable ELB Stickiness
 - Enable SSL Termination
 - Enable ELB Health Checks
 - Enable Cross Zone Load Balancing

Answer c - Enabling ELB Health Checks, the ELB will not send traffic to unhealthy (crashed) EC2 Instances

11. You are working as a Solutions Architect for a company and you are required to design an architecture for a high-performance, low-latency application that will receive millions of requests per second. Which type of Elastic Load Balancer should you choose?
- Application Load Balancer
 - Classic Load Balancer
 - Network Load Balancer

Answer c. - Network Load Balancer provides the highest performance and lowest latency if the application needs it

12. Application Load Balancers support the following protocols, BUT
- a. HTTP
 - b. HTTPS
 - c. WebSocket
 - d. TCP

Answer d. TCP - Network Load Balancers support both TCP and UDP Protocol

13. Application Load Balancers can route traffic to different Target Groups based on the following BUT
- a. Hostname
 - b. Client's Location (Geography)
 - c. Request URL Path
 - d. Source IP Address

Answer (b) - ALBs can route traffic to Target Groups based on URL Path, Hostname, HTTP Headers, and Query Strings

14. Registered targets in a Target Groups for an Application Load Balancer can be one of the following, EXCEPT:
- a. Network Load Balancer
 - b. Lambda Functions
 - c. Private IP Address
 - d. EC2 Instances

Answer (a) - ALB Target Group can include the following - EC2 Instances, Lambda Functions, or private IP Addresses

15. For compliance purposes, you would like to expose a fixed static IP address to your end-users so that they can write firewall rules that will be stable and approved by regulators. What type of Elastic Load Balancer would you choose?
- a. Application Load Balancer with an Elastic IP attached to it
 - b. Network Load Balancer
 - c. Classic Load Balancer

Answer (b) Network Load Balancer - Network Load Balancer has one static IP address per AZ and can attach an Elastic IP Address.

16. You want to create a custom application-based cookie in your Application Load Balancer. Which of the following you can use as a cookie name?

- a. AWSALBAPP
- b. AWSALBTG
- c. APPUSERC
- d. AWSALB

Answer c - Through PoE, c is the best option for the given question and condition.

17. You have a Network Load Balancer that distributes traffic across a set of EC2 instances in us-east-1. You have 2 EC2 instances in us-east-1b AZ and 5 EC2 instances in us-east-1e AZ. You have noticed that the CPU utilization is higher in the EC2 instances in us-east-1b AZ. After more investigation, you noticed that the traffic is equally distributed across the two AZs. How would you solve this problem?

- a. Enable Sticky Session
- b. Enable Cross Zone Load Balancing
- c. Enable ELB Health Checks
- d. Enable SSL Termination

Answer b - Recall that Cross Zone Load Balancing, each traffic is distributed evenly across all registered EC2 Instances in all AZs

18. You have an Application Load Balancer that is configured to redirect traffic to 3 Target Groups based on the following hostnames: users.example.com, api.external.example.com, and checkout.example.com. You would like to configure HTTPS for each of these hostnames. How do you configure the ALB to make this work?

- a. Use an HTTP to HTTPS redirect rule
- b. Use a security group SSL certificate
- c. Use Server Name Indication (SNI)

Answer c - SNI allows developers to expose multiple HTTPS applications each with its own SSL certificate on the same listener.

19. You have an application hosted on a set of EC2 instances managed by an Auto Scaling Group that you configured both desired and maximum capacity to 3. Also, you have created a CloudWatch Alarm that is configured to scale out your ASG when CPU Utilization reaches 60%. Your application suddenly received huge traffic and is now running at 80% CPU Utilization. What will happen?

- a. The desired capacity will go up to 4 and the maximum capacity will stay at 3

- b. The desired capacity will go up to 4 and the maximum capacity will stay at 4
- c. Nothing

Answer c - Nothing as the ASG cannot go over the maximum capacity configured during scale out events.

20. You have an Auto Scaling Group fronted by an Application Load Balancer. You have configured the ASG to use ALB Health Checks, then one EC2 instance has just been reported unhealthy. What will happen to the EC2 instance?

- a. The ASG will keep the instance running and restart the application
- b. The ASG will detach the EC2 Instance and leave it running
- c. The ASG will terminate the EC2 Instance

Answer c - When an EC2 instance fails at ALB Health Checks, it is marked unhealthy and will be terminated while the ASG launches a new EC2 Instance

21. Your boss asked you to scale your Auto Scaling Group based on the number of requests per minute your application makes to your database. What should you do?

- a. Tell the boss that it is impossible
- b. Create a CloudWatch custom metric then create a CloudWatch alarms on this metric to scale the ASG
- c. Enable Detailed Monitoring then create a CloudWatch Alarm to scale the ASG

Answer b - Since no CloudWatch alarm metric exists, developers will need to create a CloudWatch Custom Metric then create the alarm.

22. Your application hosted on EC2 instances managed by an Auto Scaling Group suddenly receives a spike in traffic which triggers your ASG to scale out and a new EC2 instance has been launched. The traffic continuously increases but the ASG doesn't launch any new EC2 instances immediately but after 5 minutes. What is a possible cause for this behavior?

- a. Cooldown Period
- b. Lifecycle Hooks
- c. Target Tracking Policy
- d. Launch Template

Answer a. Cooldown Period - After each scaling activity, a cooldown period occurs in which the ASG does not launch or terminate EC2 instance.

23. You're planning for a new solution that requires a MySQL database that must be available even in case of a disaster in one of the Availability Zones. What should you use?

- a. Create Read Replicas
- b. Enable Encryption
- c. Enable Multi-AZ

Answer d. Multi-AZ helps when you plan a disaster recovery for an entire AZ going down.

24. We have an RDS database that struggles to keep up with the demand of requests from our website. Our million users mostly read news, and we don't post news very often. Which solution is NOT adapted to this problem?

- a. ElastiCache Cluster
- b. RDS Multi AZ
- c. RDS Read Replicas

Answer b. RDS Multi-AZ - The question is asking which solution is NOT adapted to this problem.

25. Your application running on a fleet of EC2 instances managed by an Auto Scaling Group behind an Application Load Balancer. Users have to constantly log back in and you don't want to enable Sticky Sessions on your ALB as you fear it will overload some EC2 instances. What should you do?

- a. Use your own custom Load Balancer on EC2 Instances instead of using ALB
- b. Store session data in RDS
- c. Store session data in ElastiCache
- d. Store session data in a shared EBS volume

Answer c. Storing session data in ElastiCache is a common pattern to ensure different EC2 instances can retrieve your user's state if needed.

26. An analytics application is currently performing its queries against your main production RDS database. These queries run at any time of the day and slow down the RDS database which impacts your users' experience. What should you do to improve the users' experience?

- a. Setup a Read Replica
- b. Setup Multi-AZ

- c. Run the analytics queries at night

Answer a. - Read Replicas will help as the analytics application can now perform queries against it and will not impact the main production RDS db.

27. Your company has a production Node.js application that is using RDS MySQL 5.6 as its database. A new application programmed in Java will perform some heavy analytics workload to create a dashboard on a regular hourly basis. What is the most cost-effective solution you can implement to minimize disruption for the main application?

- a. Enable Multi AZ for the RDS database and run the analytics workload on the standby database
- b. Create a Read Replica in a different AZ and run the analytics workload on the replica database
- c. Create a Read Replica in a different AD and run the analytics workload on the source database

Answer b. Creating a Read Replica in a different AZ allows the analytics workload to run without impacting the performance of your primary database, ensuring that your production application remains unaffected. This approach is also cost-effective, as it leverages the replication feature of RDS for read-heavy operations like analytics.

28. You would like to create a disaster recovery strategy for your RDS PostgreSQL database so that in case of a regional outage the database can be quickly made available for both read and write workloads in another AWS Region. The DR database must be highly available. What do you recommend?

- a. Create a Read Replica in the same region and enable Multi-AZ on the main database
- b. Create a Read Replica in the same region and enable Multi-AZ on the Read Replica
- c. Create a Read Replica in a different region and enable Multi-AZ on the Read Replica
- d. Enable Multi-Region option on the main database

Answer c. A Read Replica in a different region can be used as a standby database and promoted to become the new production database in case of a regional disruption, Thus, having a highly available (because of Multi-AZ) RDS DB Instance in the destination AWS Region with both read and write available.

29. For your RDS database, you can have up to Read Replicas.

Answer 15

30. YES or NO - You have an un-encrypted RDS DB instance and you want to create Read Replicas. Can you configure the RDS Read Replicas to be encrypted?

Answer - NO, cannot create encrypted Read Replicas from unencrypted RDS DB Instance

31. You have purchased mycoolcompany.com on Amazon Route 53 Registrar and would like the domain to point to your Elastic Load Balancer my-elb-1234567890.us-west-2.elb.amazonaws.com. Which Route 53 Record type must you use here?

- a. CNAME
- b. Alias

Answer - Alias

32. You are developing a new application that's hosted on Elastic Beanstalk. As you are in the development process you don't mind downtime so you want your application to be deployed as soon as a new version is available. Which Elastic Beanstalk deployment option should you use?

- a. All At Once
- b. Rolling
- c. Rolling With Additional Batches
- d. Immutable

Answer a All At Once

33. A company hosting their website on AWS Elastic Beanstalk. They want a methodology to continuously release new application versions with the ability to roll back very quickly in case if there're any issues. Also, the application must be running at full capacity while releasing new versions. Which Elastic Beanstalk deployment option do you recommend?

- a. All At Once
- b. Rolling
- c. Rolling With Additional Batches
- d. Immutable

Answer d. Immutable

34. You're a DevOps engineer working for a startup company hosting their application on Elastic Beanstalk. The application is in its early phases and it has a lot of new updates every week while being used by a number of users. You want to continuously release new features without application downtime and without incurring extra costs. It's acceptable to temporarily decrease the number of running instances serving users. Which Elastic Beanstalk deployment option should you choose?

- a. All At Once
- b. Rolling
- c. Rolling With Additional Batches
- d. Immutable

Answer b. Rolling

35. Which AWS service does Elastic Beanstalk use under the hood?

- a. AWS OpsWorks
- b. AWS CloudFormation
- c. AWS Lambda
- d. Amazon EC2

Answer b. AWS CloudFormation

36. Which feature in Elastic Beanstalk allows you to automate deletions of old application versions so that new application versions can be created?

- a. Setup an .ebextensions file
- b. Use a LifeCycle Policy
- c. Define a Lambda function
- d. Use Worker Environments

Answer b. Use a LifeCycle Policy - it effectively automates the deletion of old application versions in Elastic Beanstalk allowing users to manage storage and streamline the deployment of new versions efficiently

37. You're running an application on Elastic Beanstalk. You have just finished a major update to your application. You want to deploy the new version then direct a small percentage of traffic to the new version so you can test and fall back if there're any issues. Which Elastic Beanstalk deployment option should you choose?

- a. Traffic Splitting
- b. Rolling
- c. Rolling with Additional Batches
- d. Immutable

Answer a. Traffic Splitting

38. You have been hired by a company to run some tests on their application hosted on Elastic Beanstalk. You can't run these tests on the current environment as this is the production environment, so you have to create another environment similar to the one already running. Which Elastic Beanstalk feature allows you to do this?

- a. Download the current Elastic Beanstalk environment CloudFormation template, then use it to create another environment
- b. Manually create another environment with the same configuration
- c. Elastic Beanstalk Cloning
- d. Use Worker Environment

Answer c. Elastic Beanstalk Cloning

39. An application hosted on an EC2 instance wants to upload objects to an S3 bucket using the PutObject API call, but it lacks the required permissions. What should you do?

- a. From inside the EC2 instance, run `aws configure` and insert your personal IAM credentials, because you have access to do required API call
- b. Ask an administrator to attach an IAM policy to the IAM Role on the EC2 instance that authorizes it to do the required API call
- c. Export the environment variable with your IAM credentials on the EC2 instance
- d. Use the EC2 Metadata API call

Answer b. IAM Roles are the right way to provide credentials and permissions to an EC2 instance roles

